



Islamic University of Technology
Department of Computer Science and Engineering

Lab 4: Shortest Path Algorithm

CSE 4404: Algorithms Lab
Summer 2024-25

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Task A. Shortest Path

Time Limit: 1.0 second | Memory Limit: 512 MB

You are given a weighted undirected graph with n vertices enumerated from 1 to n . Your task is to find the shortest path between vertex 1 and vertex n .

If there are multiple shortest paths, any of them will be accepted. Note that the graph may contain loops and multiple edges between the same pair of vertices.

Input Format

The first input line contains two integers n and m : the number of vertices and the number of edges ($2 \leq n \leq 10^5$, $0 \leq m \leq 10^5$).

After this, there are m lines describing the edges. Each line contains three integers a_i , b_i , and w_i : an undirected edge exists between vertex a_i and vertex b_i with weight w_i ($1 \leq a_i, b_i \leq n$, $1 \leq w_i \leq 10^6$).

Output Format

If no path exists, print -1 .

Otherwise, print the sequence of vertices representing the shortest path from vertex 1 to vertex n .

Examples

Sample Input	Sample Output
5 6 1 2 2 2 5 5 2 3 4 1 4 1 4 3 3 3 5 1	1 4 3 5
3 1 1 2 5	-1

Task B. The Escape

Time Limit: 1.0 second | Memory Limit: 512 MB

A secret operative is trapped in a city under heavy surveillance. The city is modelled as a network of n checkpoints (numbered 0 to $n-1$) connected by m monitored routes. Intelligence analysts have assessed each route and assigned it a probability of being *clear* — free of enemy patrols at the time of the escape.

The operative must travel from checkpoint *start* to the extraction point *end*. Since every route on the chosen path must be clear for the mission to succeed, the overall probability of a successful escape is the **product** of the clearance probabilities along the path.

Help the operative find the route that maximises the chance of reaching the extraction point alive. If no route exists between the two checkpoints, the probability of escape is 0.

Input Format

The first input line contains two integers n and m : the number of checkpoints and the number of monitored routes ($2 \leq n \leq 10^4$, $0 \leq m \leq 2 \cdot 10^4$).

The next m lines each contain two integers u , v and a floating-point number w : an undirected route between checkpoint u and checkpoint v with clearance probability w ($0 \leq u, v < n$, $0 \leq w \leq 1$).

The final line contains two integers *start* and *end*: the operative’s current checkpoint and the extraction point ($0 \leq start, end < n$).

Output Format

Print a single floating-point number — the maximum probability of a successful escape from *start* to *end*.

The answer is considered correct if its absolute or relative error does not exceed 10^{-5} .

Examples

Sample Input	Sample Output
3 3 0 1 0.5 1 2 0.5 0 2 0.2 0 2	0.25000

Notes

In the sample, two escape routes exist from checkpoint 0 to checkpoint 2: the direct route $0 \rightarrow 2$ with clearance probability 0.2, and the route $0 \rightarrow 1 \rightarrow 2$ with probability $0.5 \times 0.5 = 0.25$. The operative should take the longer route — the maximum escape probability is 0.25.

Task C. Second Best Route

Time Limit: 2 seconds | Memory Limit: 256 MB

A traveler has already found the fastest route from city 1 to city N . Now they want an alternative — a route that is **strictly longer** than the shortest, but as short as possible among all such alternatives.

There are N cities connected by M bidirectional roads. The i -th road connects city u_i and city v_i with a travel time of x_i . A path may revisit cities and roads any number of times.

Find the length of the second shortest path from city 1 to city N — the shortest path whose total travel time is strictly greater than the shortest path.

Input Format

The first line contains two integers N and M ($2 \leq N \leq 10^5$, $1 \leq M \leq 3 \times 10^5$).

Each of the next M lines contains three integers u_i , v_i and x_i ($1 \leq u_i, v_i \leq N$, $u_i \neq v_i$, $1 \leq x_i \leq 10^9$) describing a bidirectional road.

There may be multiple roads between the same pair of cities.

Output Format

If no path exists from city 1 to city N , print -1 .

Otherwise, output a single integer — the length of the second shortest path from city 1 to city N .

Examples

Sample Input	Sample Output
4 4 1 2 1 2 3 2 3 4 3 1 3 7	8
5 6 1 2 3 1 3 2 2 3 1 3 4 2 2 4 8 4 5 1	7
4 2 1 2 3 3 4 5	-1

Notes

In the first sample, the shortest path from 1 to 4 is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ with total travel time $1 + 2 + 3 = 6$. The second shortest path is $1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ with total travel time $1 + 1 + 1 + 2 + 3 = 8$.

In the second sample, the shortest path from 1 to 5 is $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$ with total travel time $2 + 2 + 1 = 5$. The second shortest is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$ with total travel time $3 + 1 + 2 + 1 = 7$.

In the third sample, city 4 is unreachable from city 1, so the answer is -1 .

Marks Distribution

Task	Marks
Task A	33%
Task B	33%
Task C	34%

Practice Problems

Problem	Links
Shortest Routes I	CSES: https://cses.fi/problemset/task/1671
Jzzhu and Train Routes	Codeforces: https://codeforces.com/problemset/problem/449/B
Dijkstra?	Codeforces: https://codeforces.com/problemset/problem/20/C
Sending Email	UVA: https://onlinejudge.org/index.php?option=onlinejudge&Itemid=8&